TOPIC: STUDENT PERFORMANCE PREDICTION USING MACHINE LEARNING

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INTRODUCTION

- MACHINE LEARNING (ML) MEANS TEACHING COMPUTERS TO LEARN FROM DATA.
- IN EDUCATION, ML HELPS TEACHERS UNDERSTAND HOW WELL STUDENTS MAY PERFORM.
- PREDICTING PERFORMANCE HELPS IMPROVE RESULTS AND GIVE SUPPORT TO WEAK STUDENTS.

WHAT IS STUDENT PERFORMANCE PREDICTION?

IT IS A SYSTEM THAT PREDICTS HOW A STUDENT WILL PERFORM (PASS/FAIL, MARKS, GRADES).

IT USES PAST DATA LIKE:

- ATTENDANCE
- ASSIGNMENT MARKS
- STUDY TIME
- PREVIOUS EXAM SCORES
- BEHAVIOUR/PARTICIPATION

ML LEARNS PATTERNS FROM THIS DATA AND GIVES PREDICTIONS.

WHY IS PREDICTION IMPORTANT?

- HELPS IDENTIFY WEAK STUDENTS EARLY
- HELPS TEACHERS GIVE EXTRA SUPPORT
- HELPS IMPROVE TEACHING STRATEGIES
- STUDENTS CAN UNDERSTAND THEIR WEAK AREAS

SCHOOLS CAN IMPROVE OVERALL PERFORMANCE

HOW MACHINE LEARNING WORKS HERE

Collect Data:-(attendance, marks, study hours etc.)

Clean the Data:-(remove errors, fill missing values)

Choose ML Algorithm:-

Example:

Linear Regression

Decision Tree

Random Forest

SVM

Train the Model:-

Model learns patterns from data.

- Test the Model:-
- Check accuracy on new data.
- Make Predictions:-
- Predict students' future marks or performance.

COMMON ALGORITHMS USED

1. LINEAR REGRESSION

PREDICTS CONTINUOUS MARKS.

2. DECISION TREE

PREDICTS PASS/FAIL BASED ON CONDITIONS.

3. RANDOM FOREST

MORE ACCURATE, USES MANY TREES.

4. K-NEAREST NEIGHBORS (KNN)

PREDICTS BASED ON STUDENTS SIMILAR TO YOU.

EXAMPLE PREDICTION

- Example Prediction
- Input Data (features):
- Attendance – 85%
- Study Hours – 3 hours/day
- Assignment Score – 80%
- ML Model Output:
- Predicted Marks: 82%
- or
- Result: PASS

ADVANTAGES

- Saves teacher time
- Fast and accurate predictions
- Better decision making
- Early warning for low-performing students

LIMITATIONS

- Needs good-quality data
- Prediction is not 100% accurate
- Students may improve or worsen unexpectedly
- Sensitive data must be protected

REAL LIFE APPLICATIONS

- Real-Life Applications
- Schools tracking student progress
- Online learning apps like Byju's, Unacademy
- Universities planning interventions
- Predicting dropout risks

CONCLUSION

- ML helps understand student performance easily
- Supports teachers and students
- Helpful tool for better learning outcomes
- Future: More smart education systems with AI

THANK YOU